

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and

network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds

using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid

topology combines the best elements of multiple designs: a fiberbased ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is

connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation

Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution
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Q3) Question 3 – Bus Topology for a Small Classroom Lab

Problem Statement

A small classroom laboratory requires a simple and economical computer network. The laboratory has a limited budget, so the network should use minimum networking hardware while still allowing students to communicate and share resources.

The network must support:

- Multiple student computers
- Basic file sharing
- Internet browsing
- Communication between computers
- Low installation cost
- Simple network configuration
- Easy resource sharing
- Suitable performance for light network usage

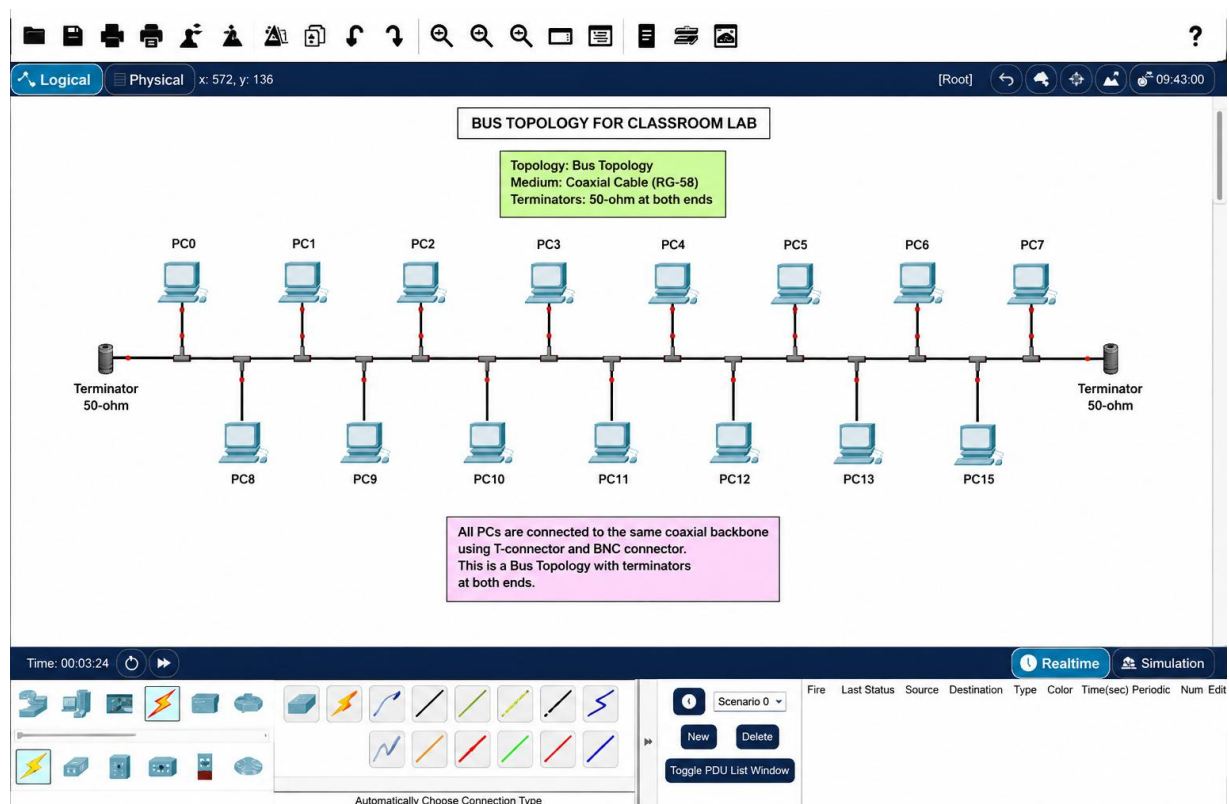
The major challenge is to design a network that provides basic connectivity while keeping the **hardware and installation cost as low as possible**.

Proposed Solution

The proposed solution is a **Bus Topology**.

In a bus topology, all computers are connected to a **single common communication line called the bus or backbone**. This eliminates the requirement for a central switch or hub in the traditional bus arrangement and reduces the overall hardware cost.

The network structure is:



50Ω Terminator

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PC0

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PC1

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PC2

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PC3

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PC4

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PC5

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50Ω Terminator

Common Bus Cable

All computers share the same communication medium.

5. Implementation of Bus Topology

The proposed classroom network can be implemented using the following components:

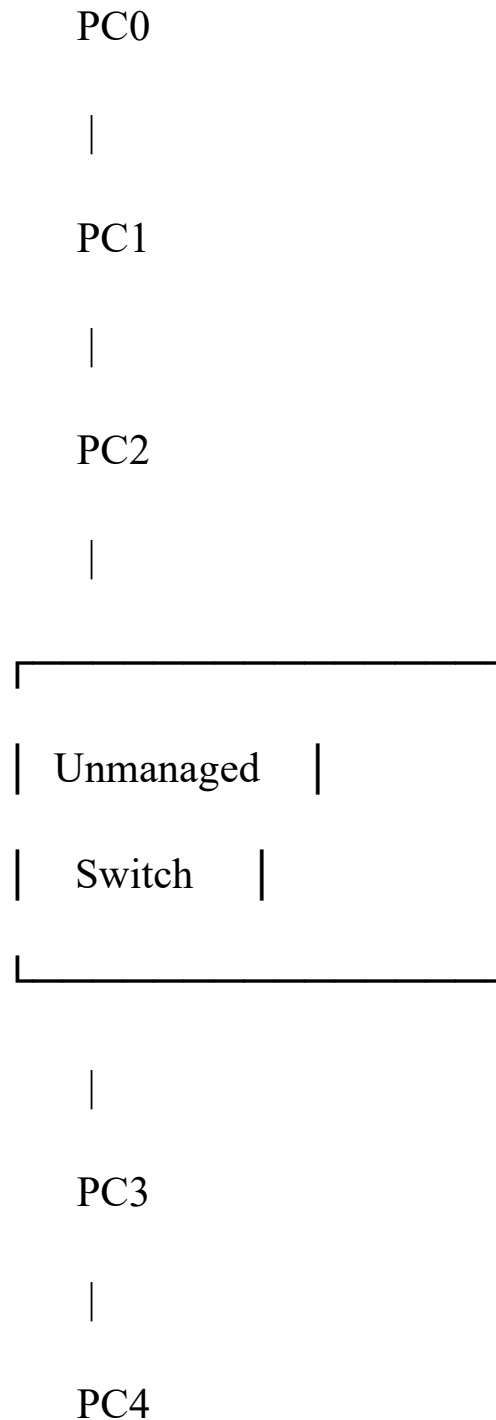
- **Computers** – Used by students for learning and accessing network resources.
- **RG-58 Coaxial Cable** – Used as the main communication backbone.
- **T-Connectors** – Used to connect individual computers to the backbone.
- **BNC Connectors** – Used for connecting the coaxial cable to network devices.
- **50-ohm Terminators** – Placed at both ends of the cable to prevent signal reflection.
- **Network Interface Cards** – Allow computers to communicate through the network.

In a traditional bus topology, a single **RG-58, 50-ohm coaxial cable** runs through the classroom, and each computer is connected to the backbone using a T-connector and BNC connector. Terminators are placed at both ends of the cable.

6. Modern Low-Cost Implementation

For a modern classroom environment, a low-cost implementation can use an **8-port or 16-port unmanaged switch** along with **Cat5e/Cat6 UTP cables**.

The arrangement can be represented as:



The inexpensive switch provides connectivity to the computers while keeping the implementation simple and affordable. The given problem specifically identifies this as a contemporary low-cost alternative.

7. Working of the Network

In a bus topology, all computers share the same communication channel. When a computer sends data, the signal travels through the common backbone.

For example:

PC0 → Common Bus → PC1

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PC2 / PC3

The intended computer receives the transmitted information while the other computers ignore data that is not addressed to them.

The classroom network can be used for:

- Internet browsing
- File sharing
- Sharing educational materials
- Accessing common resources
- Basic network communication

A dedicated server is not required. The teacher's computer can share files and other resources using the built-in file-sharing facilities available in Windows or Linux.

8. Problems and Solutions

Problem 1: High Network Installation Cost

A classroom with a limited budget cannot afford expensive enterprise networking equipment.

Solution

Bus topology reduces hardware requirements because all computers share a common communication line. Therefore, a central switch or hub is not required in the traditional implementation.

Limited Budget



Less Hardware Required



Lower Installation Cost



Bus Topology

This makes bus topology suitable for a small classroom laboratory.

Problem 2: Need for Simple Network Design

A classroom laboratory requires a simple network that can be installed without complex infrastructure.

Solution

In a bus topology, all computers are connected to one common backbone cable.

PC0 — PC1 — PC2 — PC3 — PC4

Common Communication Bus

The simple structure makes it suitable for a small and controlled classroom environment.

Problem 3: Network Collisions

Since all computers share the same communication medium, multiple computers may attempt to transmit data at the same time.

Solution

The network should be used for **light network traffic**, such as basic internet browsing and file sharing. In a supervised classroom environment, this limitation can be acceptable.

Multiple Computers



Shared Bus



Possible Collisions



Keep Network Traffic Low

Problem 4: Backbone Cable Failure

A major disadvantage of bus topology is that a failure in the main communication cable can affect all connected computers.

Solution

The backbone cable should be properly installed and regularly checked.

Backbone Cable Failure



Communication Interrupted



Check Cable and Connectors



Replace Damaged Section



Restore Network

The source identifies a single cable break as a major limitation because it can disrupt all connected machines.

Problem 5: Need for Resource Sharing

Students may need to share files and educational resources between computers.

Solution

A dedicated server is not necessary for a small classroom. The teacher's PC can provide file-sharing services using Windows or Linux built-in networking facilities.

Teacher PC

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Shared Files / Resources

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Student Computers

9. Advantages of Bus Topology

Bus topology provides several advantages for a small classroom laboratory:

- **Low hardware cost**
- **Simple network structure**
- **Less networking equipment**
- **Suitable for small networks**
- **Easy to understand and implement**
- **Suitable for light network traffic**
- **No dedicated server required**
- **Cost-effective for a limited-budget environment**

The main reason for selecting this topology is its economical design and suitability for a small classroom with light network usage.

10. Limitations of Bus Topology

Although bus topology is inexpensive, it has some limitations:

- A failure in the main cable can affect the entire network.
- Collisions may occur because all devices share the same communication medium.
- It is not suitable for large networks.
- Troubleshooting a backbone failure can be difficult.

- Network performance may decrease when network traffic increases.

These limitations make bus topology more appropriate for **small and supervised environments** rather than large enterprise networks.

11. Why Bus Topology Was Selected

Bus topology was selected because the problem specifically describes a **small classroom laboratory with a tight budget**.

It provides:

- Low installation cost
- Simple implementation
- Minimum hardware requirements
- Basic file-sharing capability
- Support for light network usage
- Easy implementation in a small classroom

The trade-off between low cost and limitations such as collisions and cable failure is acceptable for a supervised classroom environment.

12. Testing of the Network

After implementing the network, connectivity between computers should be tested.

For example, the teacher can check whether one computer can communicate with another computer.

PC0



Send Test Data



Common Bus



PC1



Check Response

File-sharing can also be tested by sharing a sample file from the teacher's computer and accessing it from a student computer.

The following tests can be performed:

1. Check physical cable connections.
2. Check the network interface of each computer.
3. Test communication between computers.
4. Test file sharing.

5. Test internet connectivity if internet access is provided.
6. Check the network after disconnecting and reconnecting a cable.

Successful communication confirms that the network is operating correctly.

13. Final Conclusion

A **Bus Topology** was designed for a small classroom laboratory with a limited budget. All computers share a common communication backbone, reducing the amount of networking hardware required. A traditional implementation can use **RG-58 coaxial cable, T-connectors, BNC connectors and 50-ohm terminators**. A modern low-cost implementation can use an inexpensive unmanaged switch with Cat5e/Cat6 cables.

The main advantages of the design are **low cost, simple implementation and suitability for small networks**. However, the network has limitations such as collisions and the possibility that a backbone cable failure can disrupt all connected computers.

Therefore, **bus topology is a suitable and economical solution for a small, supervised classroom laboratory where network usage is relatively light and the primary requirement is to minimize cost while providing basic communication and resource sharing.**